THE SOLUTIONS

1. (a) Three vectors in $\mathbb{R}^3$ are given as $\underline{\mathbf{a}} = (0,1,3)^T$, $\underline{\mathbf{b}} = (1,2,4)^T$, and $\underline{\mathbf{c}} = (1,-1,0)^T$.

(i) A plane P_1 through the origin is obtained by taking all linear combinations of $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$. A plane P_2 is obtained by adding $\underline{\mathbf{c}}$ to every point in P_1. Obtain the cartesian equation of plane P_2. [4]

(ii) Use projection to find the distance from the point $M(3,-5,-2)$ to the plane P_2. Find the vector equation of the line L_1 through M, orthogonal to P_2. Finally, obtain the point of intersection N of this line with P_2, and the magnitude of the vector $\overrightarrow{MN}$. Comment on this! [5]

Solution: (i) The normal vector for P_1 can be found with the cross product $\underline{\mathbf{b}} \times \underline{\mathbf{a}} = \underline{\mathbf{n}} = (2,-3,1)^T$. Hence the cartesian equation of P_1 is $2x-3y+z=0$. As the origin is on P_1, this is translated to $(1,-1,0)$ by adding $\underline{\mathbf{c}}$. So we need a constand D such that P_2 has equation $2x-3y+z=D$. Taking $(1,-1,0)$ as a point on P_2, we substitute these values and obtain $D=5$. Hence the cartesian equation of P_2 is $2x-3y+z=5$.

Of those that understood the link between linear combinations of two vectors and a plane, most got P_1 right. Many tried various other methods, such as taking the vectors as three points on the plane, leading nowhere. Of those that found P_1, many had trouble working out the correct way to translate it to P_2.

(ii) We use the known point $(1,-1,0)$ on P_2 to calculate a vector $\underline{\mathbf{x}} = (-2,4,2)^T$ from M to the plane. The projection of this vector onto the normal vector of the plane $\underline{\mathbf{n}} = (2,-3,1)^T$ is given by $\lambda\underline{\mathbf{n}}$ where

$$\lambda = \frac{\underline{\mathbf{n}} \cdot \underline{\mathbf{x}}}{\underline{\mathbf{n}} \cdot \underline{\mathbf{n}}} = \frac{-4-12+2}{4+9+1} = -1$$

so the projection is the vector $-\underline{\mathbf{n}} = (-2,3,-1)^T$. The magnitude of this vector is the distance from M to the plane: $\sqrt{14}$. Line L_1 has vector equation $(x,y,z)^T = (3,-5,-2)^T + p(2,-3,1)^T$, for any real p. To find its intersection with P_1 substitute $x = 3+2p, y = -5-3p, z = -2+p$ into the equation of the plane:

$$2(3+2p) - 3(-5-3p) + (-2+p) = 5 \Rightarrow p = -1$$

Hence the intersection of plane and line is the point N on the line: $(3,-5,-2) - (2,-3,1) = (1,-2,-3)$. This is also the point on P_2 nearest to M, so $\overrightarrow{MN} = (-2,3,-1)^T = -\underline{\mathbf{n}}$, and the magnitude $|\overrightarrow{MN}| = \sqrt{14}$, as calculated with projection.

Most people got one of the two, projection or intersection, but not both.

(b) A double integral is given as

$$I \;=\; \int_{x=0}^{\pi/2}\int_{y=x}^{\pi/2} \cos\left(\frac{2y^2}{\pi}\right)\, dy\, dx\,.$$

To evaluate this integral, first sketch the region of integration R, then change the order of integration, and finally obtain the value of I. [6]

Solution: We can't integrate w.r.t. y, so need to identify R in the first quadrant, enclosed by the $y-$axis, the horizontal line $y=\pi/2$, and the diagonal $y=x$. Sketch at end. To change the order of integration, see sketch, we begin with x, where R has $x=0\ldots y$, followed by $y=0\ldots\pi/2$:

$$I=\int_{y=0}^{\pi/2}\left(\int_{x=0}^{y}\cos\left(\frac{2y^2}{\pi}\right)\,dx\right)\,dy \;=\; \int_{y=0}^{\pi/2}\left[x\cos\left(\frac{2y^2}{\pi}\right)\right]_{x=0}^{y}\,dy \;=\; \int_{y=0}^{\pi/2} y\cos\left(\frac{2y^2}{\pi}\right)\,dy$$

Let $u=\dfrac{2y^2}{\pi}\Rightarrow du=\dfrac{4y}{\pi}\,dy$, and $y=0\ldots\pi/2\Rightarrow u=0\ldots\pi/2$, so we rewrite as

$$I=\frac{\pi}{4}\int_0^{\pi/2}\cos u\,du \;=\; \frac{\pi}{4}\left[\sin u\right]_0^{\pi/2} \;=\; \frac{\pi}{4}\,.$$

Changing the order of integration led to a lot of incorrect limits in the integrals, even when the sketch of R was correct. Quite a large minority had variable limits in the outer integral.

(c) A function of two variables is given as

$$f(x,y) \;=\; (x^2-1+y)(x^2-1-y)\,.$$

(i) Obtain all stationary points of $f(x,y)$ and use the Hessian determinant to classify them as maximum, minimum or saddle point. [5]

(ii) Sketch the zero contour of $f(x,y)$. Separately, use the zero contour and the information from (i), to sketch the full contour plot. Be sure to explain your reasoning. [5]

Solution: (i) Multiply out to obtain $f(x,y) \;=\; x^4-2x^2+1-y^2$. Obtain the first partial derivatives and set these equal to zero for the stationary points:

$$(I)\; f_x \;=\; 4x^3-4x \;=\; 0\,, \quad \text{and} \quad (II)\; f_y \;=\; -2y \;=\; 0\,.$$

From the first equation we get $x=0,\pm1$, from the second $y=0$ so we have three stationary points: the origin and

$$P_1(1,0)\,, P_2(-1,0)\,.$$

Now find the Hessian determinant:

$$D \;=\; f_{xx}f_{yy}-(f_{xy})^2 \;=\; (12x^2-4)(-2)-(0)^2 \;=\; 8(1-3x^2)\,.$$

Evaluating at $P_{1,2}$ we have $D=-16<0$ so these are saddle points. At the origin, we have $D=8>0$ so it's an extremum. Here, $f_{yy}=-2$, so the origin is a local maximum.

Mostly done well.

(ii) The zero contour is obtained by setting

$$f(x,y) \;=\; (x^2-1+y)(x^2-1-y) \;=\; 0 \Rightarrow y = 1-x^2 \quad \text{or} \quad y = x^2-1\,,$$

so the zero contour consists of two parabolae, intersecting at $(\pm 1, 0)$, the saddle points as expected. This allows us to draw the rest of the contour plot, beginning with the area, A (sketch at end), enclosed by the parabolae, with the origin, a maximum at its centre. On the zero contour the function is equal to zero, so everywhere inside A, we must have $f > 0$, moving towards the maximum at the origin. Moving down from the origin towards area B, we cross the zero contour, and with f decreasing, we must have $f < 0$ in area B, with contours further away getting larger and larger in the negative sense. The same holds, by symmetry, for area C. Finally begin with any contour in A, where $f < 0$,and move to the right, crossing the zero contour, into area D, where we have $f > 0$, with contours further away getting larger in the positive sense. The same holds, by symmetry, for area E, completing the contour plot.

Very few people read the question correctly, and did not sketch the zero-contour separately. Equally few people explained their reasoning, even when the contour plot was correct.

2. (a) Obtain the solution of the first-order Ordinary Differential Equation

$$x\sin t - t - x^2 e^{2t} \;=\; \left[\cos t + \sin x + x e^{2t}\,\right] \frac{dx}{dt}$$

which satifisies the initial condition $x(0) = 1$. [6]

Solution: Rearrange the ODE to

$$x^2 e^{2t} - x\sin t + t + \left[\cos t + \sin x + x e^{2t}\,\right] \frac{dx}{dt} \;=\; 0\,,$$

and identify $P(t,x) = x^2 e^{2t} - x\sin t + t$, and $Q(t,x) = \cos t + \sin x + x e^{2t}$, as in the formula sheet. Test if this is an exact ODE:

$$\frac{\partial P}{\partial x} \;=\; 2x e^{2t} - \sin t \quad \text{and} \quad \frac{\partial Q}{\partial t} \;=\; -\sin t + 2x e^{2t}\,,$$

are equal, so it is exact, and a function $f(t,x)$ exists with

$$\frac{\partial f}{\partial t} \;=\; P(t,x) \;=\; x^2 e^{2t} - x\sin t + t\,, \quad \text{and} \quad \frac{\partial f}{\partial x} \;=\; Q(t,x) \;=\; \cos t + \sin x + x e^{2t}$$

and we can integrate to find f:

$$f(t,x) \;=\; \int P\,dt \;=\; \int x^2 e^{2t} - x\sin t + t\;dt \;=\; \frac{1}{2}x^2 e^{2t} + x\cos t + \frac{1}{2}t^2 + m(x)\,,$$

and

$$f(t,x) \;=\; \int Q\,dx \;=\; \int \cos t + \sin x + x e^{2t}\;dx \;=\; x\cos t - \cos x + \frac{1}{2}x^2 e^{2t} + n(t)\,,$$

where $m(x)$ and $n(t)$ are arbitrary functions. Comparing the two expressions, we see that $n(t) = \frac{1}{2}t^2$, and $m(x) = -\cos x$ and the solution is given by $f(t,x) = constant$:

$$\frac{1}{2}x^2 e^{2t} + x\cos t + \frac{1}{2}t^2 - \cos x \;=\; C\,,$$

and we apply the initial condition $x(0) = 1$ to find $C = \frac{3}{2} - \cos 1$.

Mostly done well. A lot of people didn't check that the equation is exact. A few did test for exactness, but compared P_t and Q_x, which are not the same, and so got stuck. Many had trouble finding the correct constant.

(b) Let a function be defined as $f(x) = \begin{cases} 0 & \text{for} \quad 0 \le x < \pi/2 \,, \\ \pi - x & \text{for} \quad \pi/2 \le x < \pi \,. \end{cases}$

Let $g(x)$ be the even extension, $h(x)$ the odd extension of $f(x)$, both with period $T = 2\pi$.

(i) Sketch $g(x)$ and $h(x)$, for $-3\pi \le x \le 3\pi$. If Fourier series for $g(x)$ and $h(x)$ are calculated (do not do this yet!) what is the value each Fourier series takes at $x = -3\pi/2$? State the rate of decay of the Fourier coefficients for each series. [6]

(ii) Now obtain the first six non-zero terms of the Fourier series for $h(x)$. [8]

(iii) Use an appropriate choice of x in (ii), to find $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$ [5]

Solution: (i) Sketch at end. Both extensions have a discontinuity at $x = -3\pi/2$ so the series converges to the average of the values at either end:

$$\frac{1}{2}\left(g(x_-) + g(x_+)\right) = \frac{1}{2}(\pi/2 + 0) = \frac{\pi}{4} \,,$$

and the same for $h(x)$. The presence of the discontinuity in both functions means that the Fourier coefficients decay as $1/n$.

Many incorrect sketches. Many errors in values at discontinuities, and many didn't mention rates of decay.

(ii) The function $h(x)$ is odd, so $a_n = 0$. Now for $T = 2\pi$,

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} h(x)\sin(2n\pi x/T)\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} h(x)\sin(nx)\,dx = \frac{2}{\pi}\int_0^{\pi} h(x)\sin(nx)\,dx = \frac{2}{\pi}\int_{\pi/2}^{\pi} (\pi - x)\sin(nx)\,dx$$

using half-range approach. Hence, integrating by parts, we have $b_n =$

$$\frac{2}{\pi}\left[-(\pi - x)\frac{\cos(nx)}{n}\right]_{\pi/2}^{\pi} - \frac{2}{n\pi}\int_{\pi/2}^{\pi}\cos(nx)\,dx = \frac{1}{n}\cos\left(\frac{n\pi}{2}\right) - \left[\frac{2}{n^2\pi}\sin(nx)\right]_{\pi/2}^{\pi} = \frac{1}{n}\cos\left(\frac{n\pi}{2}\right) + \frac{2}{n^2\pi}\sin\left(\frac{n\pi}{2}\right) \,. (*)$$

We note that $\cos(n\pi/2)$ is zero when n is odd, and ± 1 when n is even, and vice-versa for $\sin(n\pi/2)$. Begin with n odd:

$$b_1 \;=\; \frac{2}{\pi}\sin(\pi/2) = \frac{2}{\pi} \,, \quad b_3 \;=\; \frac{2}{3^2\pi}\sin(3\pi/2) = -\frac{2}{9\pi} \,, \quad b_5 \;=\; \frac{2}{25\pi} \,.$$

Likewise, for even n:

$$b_2 \;=\; \frac{1}{2}\cos(\pi) = -\frac{1}{2} \,, \quad b_4 \;=\; \frac{1}{4}\cos(2\pi) = \frac{1}{4} \,, \quad b_6 = -\frac{1}{6}$$

and the Fourier series is therefore

$$h(x) \;=\; \frac{2}{\pi}\sin(x) - \frac{1}{2}\sin(2x) - \frac{2}{9\pi}\sin(3x) + \frac{1}{4}\sin(4x) + \frac{2}{25\pi}\sin(5x) - \frac{1}{6}\sin(6x) + \cdots$$

A lot of false starts. Integrating from $-\pi$ to π, which made everything very hard. Or integration from 0 to π, ignoring the interval where $h(x)$ is zero. Many got to $(*)$, but then got lost.

(iii) The required series is the same as $\sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^2}$, so we need a value x that will make the terms in the Fourier series with even n vanish. These have $\sin(2x), \sin(4x), \sin(6x)\ldots$ so we consider $x = \pi/2$. For this value, the odd terms have $\sin(\pi/2) = 1, \sin(3\pi/2) = -1, \sin(5\pi/2) = 1\ldots$, which gives us

$$h(\pi/2) \;=\; \frac{2}{\pi}(1) - \frac{2}{3^2\pi}(-1) + \frac{2}{5^2\pi}(1) - \frac{2}{7^2\pi}(-1)\ldots \;=\; \frac{2}{\pi}\sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^2}$$

Now consider $h(\pi/2)$. The function has a discontinuity at $x = \pi/2$, so the series converges to the average of the values on either side, which is $\pi/4$, as seen in (i). Finally, we have

$$\frac{\pi}{4} \;=\; \frac{2}{\pi}\sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^2} \Rightarrow \sum_{\substack{n=1 \\ n\ odd}}^{\infty} \frac{1}{n^2} \;=\; \frac{\pi^2}{8}\,.$$

This was doable if the answer to (ii) was correct. Quite a few got the correct value of x, but then ignored the discontinuity for h.

3. (a) A real-valued signal $f(t)$ can be written as the sum of its even and odd parts: $f(t) \;=\; f_e(t) + f_o(t)$, where $f_e(t)$ and $f_o(t)$ are even and odd functions, respectively. If $F(\omega)$ is the Fourier transform of $f(t)$, obtain the real and imaginary parts of $F(\omega)$ in terms of the Fourier transforms of $f_e(t)$ and $f_o(t)$, explaining your reasoning. [5]

Solution: Using the definition of the Fourier transform, we write

$$F(\omega) \;=\; \int_{-\infty}^{+\infty} f(t)e^{-i\omega t}\,dt \;=\; \int_{-\infty}^{+\infty} [f_e(t) + f_o(t)]e^{-i\omega t}\,dt \;=\; \int_{-\infty}^{+\infty} f_e(t)e^{-i\omega t}\,dt + \int_{-\infty}^{+\infty} f_o(t)e^{-i\omega t}\,dt$$

Hence

$$\mathscr{F}[f_e(t)] \;=\; \int_{-\infty}^{+\infty} f_e(t)e^{-i\omega t}\,dt \;=\; \int_{-\infty}^{+\infty} f_e(t)\cos(\omega t)\,dt - i\int_{-\infty}^{+\infty} f_e(t)\sin(\omega t)\,dt \;=\; \int_{-\infty}^{+\infty} f_e(t)\cos(\omega t)\,dt\,,$$

given that the symmetric integral of the odd function $f_e(t)\sin(\omega t)$ is zero. Hence $\mathscr{F}[f_e(t)]$ is purely real. Similarly,

$$\mathscr{F}[f_o(t)] \;=\; \int_{-\infty}^{+\infty} f_o(t)e^{-i\omega t}\,dt \;=\; \int_{-\infty}^{+\infty} f_o(t)\cos(\omega t)\,dt - i\int_{-\infty}^{+\infty} f_o(t)\sin(\omega t)\,dt \;=\; -i\int_{-\infty}^{+\infty} f_o(t)\sin(\omega t)\,dt\,,$$

given that the symmetric integral of the odd function $f_o(t)\cos(\omega t)$ is zero. So $\mathscr{F}[f_o(t)]$ is purely imaginary, and we have the real and imaginary parts:

$$Re[F(\omega)] = \mathscr{F}[f_e(t)] \quad and \quad Im[F(\omega)] = -i\mathscr{F}[f_o(t)]\,.$$

Not many full answers given. A lot of attempted shortcuts with no comments.

(b) Obtain $F(\omega)$, the Fourier transform of $f(t) \;=\; tH(t+1) - tH(t-1)$, where $H(t)$ is the Heaviside function, and give $F(\omega)$ in terms of trigonometric functions. [5]

Solution: We note that $f(t) = t$ for $-1 \le t \le 1$ and zero everywhere else, and this is an odd function so we use (a) to write

$$F(\omega) \;=\; \int_{-1}^{1} te^{-i\omega t}\,dt \;=\; \int_{-1}^{1} t\cos(\omega t)\,dt \;-\; i\int_{-1}^{1} t\sin(\omega t)\,dt \;=\; -2i\int_{0}^{1} t\sin(\omega t)\,dt$$

as the first integral vanishes, and the second is a symmetric integral of an even function. Integrating by parts, we have

$$F(\omega) \;=\; -2i\left[-\frac{t\cos(\omega t)}{\omega}+\frac{\sin(\omega t)}{\omega^2}\right]_0^1 \;=\; 2i\frac{\omega\cos(\omega)-\sin(\omega)}{\omega^2}\,.$$

Many people integrated $te^{-i\omega t}$ which made things difficult later.

(c) (i) Use the definition of the Laplace transform to prove the second shift theorem:
if $\mathscr{L}[f(t)] = \bar{f}(s)$, and $H(t)$ is the Heaviside function, then

$$\mathscr{L}[f(t-a)H(t-a)] = e^{-sa}\bar{f}(s) .$$

State any necessary condition. [4]

Solution:

$$\mathscr{L}[H(t-a)f(t-a)] = \int_0^\infty e^{-st}H(t-a)f(t-a)\,dt = \int_a^\infty e^{-st}f(t-a)\,dt$$

now let $\tau = t-a$ so that $d\tau = dt$, and $t = a\ldots\infty \Rightarrow \tau = 0\ldots\infty$, so we have

$$\mathscr{L}[H(t-a)f(t-a)] = \int_0^\infty e^{-s(\tau+a)}f(\tau)\,d\tau = e^{-sa}\int_0^\infty e^{-s\tau}f(\tau)\,d\tau = e^{-sa}\bar{f}(s),$$

given that τ is a dummy variable. This holds for $a \geq 0$, and if the transform $\bar{f}(s)$ exists, so we also require $\lim_{t\to\infty} f(t)e^{-st} = 0$.

Quite a few people tried to integrate from a to ∞, which let to a lot of incorrect answers, especially with $f(t-a)$. Few people gave the needed condition at the end.

(ii) Use convolution and the second shift theorem to obtain the following inverse Laplace transform:

$$\mathscr{L}^{-1}\left[\frac{e^{-2s}}{s(s^2+1)}\right] . \qquad [6]$$

Solution: First apply the second shift theorem:

$$\mathscr{L}^{-1}\left[\frac{e^{-2s}}{s(s^2+1)}\right] = \mathscr{L}^{-1}\left[\frac{1}{s(s^2+1)}\right]_{t\to t-2} H(t-2)$$

and use convolution to obtain

$$\mathscr{L}^{-1}\left[\left(\frac{1}{s}\right)\left(\frac{1}{s^2+1}\right)\right] = f*g ,$$

where $f*g$ is the convolution, and

$$f(t) = \mathscr{L}^{-1}\left[\frac{1}{s}\right] = 1 \quad \text{and} \quad g(t) = \mathscr{L}^{-1}\left[\frac{1}{s^2+1}\right] = \sin t .$$

Hence

$$\mathscr{L}^{-1}\left[\frac{1}{s(s^2+1)}\right] = \int_0^t 1\sin(u)\,du = 1-\cos t ,$$

and

$$LT^{-1}\left[\frac{e^{-2s}}{s(s^2+1)}\right] = [1-\cos(t-2)]H(t-2) .$$

Tough question: many that tried convolution got it wrong. Many people used partial fractions instead and got the correct answer. Many couldn't use the second-shift theorem correctly.

(iii) Use Laplace transforms and (ii), to solve the Ordinary Differential Equation

$$\frac{d^2x}{dt^2}+x=\begin{cases}0 & \text{for} \quad 0\le t<2\,,\\ 1 & \text{for} \quad t\ge 2\,,\end{cases}$$

which satisfies the initial conditions $x(0)=x'(0)=1$. [5]

Solution: Rewrite the ODE as

$$\frac{d^2x}{dt^2}+x \;=\; H(t-2)\,,$$

and now take transforms using the formula sheet:

$$s^2\bar{x}-sx(0)-x'(0)+\bar{x} \;=\; \frac{e^{-2s}}{s}\,,$$

Hence we have

$$(s^2+1)\bar{x} \;=\; s+1+\frac{e^{-2s}}{s} \Rightarrow \bar{x} \;=\; \frac{s+1}{s^2+1}+\frac{e^{-2s}}{s(s^2+1)}\,.$$

Using the library,

$$\mathscr{L}^{-1}\left[\frac{s+1}{s^2+1}\right] \;=\; \mathscr{L}^{-1}\left[\frac{s}{s^2+1}\right]+\mathscr{L}^{-1}\left[\frac{1}{s^2+1}\right] \;=\; \cos t+\sin t\,,$$

and combining this with the result from (ii), we get

$$x(t) \;=\; \mathscr{L}^{-1}\left[\frac{s+1}{s^2+1}+\frac{e^{-2s}}{s(s^2+1)}\right] \;=\; \cos t+\sin t+[1-\cos(t-2)]H(t-2)\,.$$

A lot of people didn't apply LT correctly to the ODE, so did not get the r.h.s seen in (ii). Of these, a lot of people did not get the e^{-2s} term, so proceeded on without the second-shift theorem.

4. (a) Two matrices are given as

$$A \;=\; \begin{pmatrix} 1 & 2 & 3\\ -1 & 1 & 1\\ \phi & 0 & 1\end{pmatrix}\,, \quad \text{and} \quad B \;=\; \begin{pmatrix} 1 & 3 & 2\\ -1 & 1 & 1\\ 2 & 1 & -1\end{pmatrix}$$

(i) Let $\phi=2$. Find $\det(A)$, as well as $\det(B)$ and $\det(AB)$. Finally, use Gauss-Jordan elimination to obtain A^{-1}. [6]

(ii) Let $\phi=3$. Give conditions on $\underline{\mathbf{b}}=(1,\theta,2)^T$ so that the matrix equation $A\underline{\mathbf{x}}=\underline{\mathbf{b}}$ has no solutions, or infinitely many solutions. Can the system have a unique solution? Either way, explain why. Finally, for the case of infinitely many solutions, find these and give a geometric interpretation. [6]

(iii) Use row operations to obtain the echelon form of B, the upper-triangular matrix U. The $LU-$decomposition of B requires the lower-triangular matrix L, so that $LU=B$. With no further calculations needed, write down L. [4]

Solution (i) Expand by third row, and

$\det(A) = \phi(2-3)+1(1+2) = -\phi+3 = 1$, and use row operations: (R_2+R_1, R_3-2R_1), to write

$$\det(B) = \begin{pmatrix} 1 & 3 & 2 \\ 0 & 4 & 3 \\ 0 & -5 & -5 \end{pmatrix} = 1(-20+15) = -5\,.$$

Hence $\det(AB)$ =$\det(A)\det(B) = 1(-5) = -5$.

For Gauss-Jordan, augment A with the identity matrix, and use row operations to make the left-half into the identity: $(A : I)$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & -1 & 0 & 0 & 1 \end{array}\right) \underset{\substack{R_2+R_1 \\ R_3-2R_1}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 3 & 4 & 1 & 1 & 0 \\ 0 & -4 & -5 & -2 & 0 & 1 \end{array}\right) \underset{\substack{ \\ -R_3 \\ -R_4}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -4 & -1 & -1 & 0 \\ 0 & 4 & 5 & 2 & 0 & -1 \end{array}\right)$$

$$\underset{R_2+R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & -1 & -1 \\ 0 & 4 & 5 & 2 & 0 & -1 \end{array}\right) \underset{\substack{R_1-2R_2 \\ R_3-4R_2}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 2 & 2 \\ 0 & 1 & 1 & 1 & -1 & -1 \\ 0 & 0 & 1 & -2 & 4 & 3 \end{array}\right) \underset{\substack{R_1-R_3 \\ R_2-R_3}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & -1 \\ 0 & 1 & 0 & 3 & -5 & -4 \\ 0 & 0 & 1 & -2 & 4 & 3 \end{array}\right)$$

so that $A^{-1} = \begin{pmatrix} 1 & -2 & -1 \\ 3 & -5 & -4 \\ -2 & 4 & 3 \end{pmatrix}$.

Some people multiplied to find AB, and then its determinant, usually incorrectly. Inverse done well.

(ii) We now have $\det(A) = 0$, making the matrix singular, so the system can never have a unique solution, as this is only possible when A is invertible. Use Gaussian elimination on the augmented coefficient matrix:

$$(A : \underline{\mathbf{b}}) = \left(\begin{array}{ccc:c} 1 & 2 & 3 & 1 \\ -1 & 1 & 1 & \theta \\ 3 & 0 & 1 & 2 \end{array}\right) \underset{\substack{R_2+R_1 \\ R_3-3R_1}}{\sim} \left(\begin{array}{ccc:c} 1 & 2 & 3 & 1 \\ 0 & 3 & 4 & \theta+1 \\ 0 & -6 & -8 & -1 \end{array}\right) \underset{R_3+2R_2}{\sim} \left(\begin{array}{ccc:c} 1 & 2 & 3 & 1 \\ 0 & 3 & 4 & \theta+1 \\ 0 & 0 & 0 & 2\theta+1 \end{array}\right)$$

and we have echelon form. If $2\theta+1 \neq 0$, there is a pivot in the right-most column of the echelon form and the system in inconsistent, so no solutions, unless $\theta = -1/2$. For $\theta = -1/2$, we have infintitely many solutions. We now have

$$(A : \underline{\mathbf{b}}) \sim \left(\begin{array}{ccc:c} 1 & 2 & 3 & 1 \\ 0 & 3 & 4 & 1/2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Row two has $3y+4z = 1/2$. Let $z = \lambda \in \mathbb{R}$, a free variable. Then $y = \dfrac{1}{6} - \dfrac{4}{3}\lambda$ and row one has $x = 1-2y-3z = 1-2\left(\dfrac{1}{6}-\dfrac{4}{3}\lambda\right)-3\lambda = \dfrac{2}{3}-\dfrac{1}{3}\lambda$. In vector form this is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/6 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1/3 \\ -4/3 \\ 1 \end{pmatrix},$$

the vector equation of a line.

Many people ignored the fact that the determinant being zero is significant here. Lots of arguments about when a solution would be unique. Gaussian elimination with θ was subject to error. After that, mostly fine.

(iii) Use row operations to obtain U, an upper triangular matrix, from B:

$$B = \begin{pmatrix} 1 & 3 & 2 \\ -1 & 1 & 1 \\ 2 & 1 & -1 \end{pmatrix} \underset{\substack{R_2+R_1 \\ R_3-2R_1}}{\sim} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 4 & 3 \\ 0 & -5 & -5 \end{pmatrix} \underset{R_3+\frac{5}{4}R_2}{\sim} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 4 & 3 \\ 0 & 0 & -5/4 \end{pmatrix} = U$$

Three row operations were used to obtain U, each can be represented as multiplication on the left by an elimination matrix:

$$(E_{32}E_{31}E_{21})A = U \Rightarrow A = (E_{21}^{-1}E_{31}^{-1}E_{32}^{-1})U = LU$$

and we can immediately write down L, applying the inverse row operations $R_3 - \frac{5}{4}R_2$, $R_3 + 2R_1$, and $R_2 - R_1$ to the identity, so

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5/4 & 1 \end{pmatrix},$$

A few people tried to get LU−decompostion of the singular matrix A, which proved difficult. Otherwise, mostly fine, for U, but lots of errors in L.

(b) Let a matrix be given as

$$A = \begin{pmatrix} -1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 2 & 1 & 4 & 5 \\ 0 & -2 & -4 & -6 \end{pmatrix}$$

Use Gaussian elimination to obtain the echelon form of A, identify rank, pivot and free columns, and hence find bases for nullspace $N(A)$ and column space $C(A)$. What does the basis for the nullspace tell us about the columns of A? Finally, obtain a particular solution of $A\underline{\mathbf{x}} = \underline{\mathbf{b}} = (7,4,-2,-8)^T$, and with no further calculation, write down the general solution. [9]

Solution: Begin with Gaussian elimination, identifying the first pivot as $a_{11} = -1$, and including $\underline{\mathbf{b}}$ from the beginning: $(A : \underline{\mathbf{b}}) =$

$$\begin{pmatrix} \boxed{-1} & 1 & 1 & 2 & : & 7 \\ 0 & 1 & 2 & 3 & : & 4 \\ 2 & 1 & 4 & 5 & : & -2 \\ 0 & -2 & -4 & -6 & : & -8 \end{pmatrix} \underset{R_3+2R_1}{\sim} \begin{pmatrix} \boxed{-1} & 1 & 1 & 2 & : & 7 \\ 0 & \boxed{1} & 2 & 3 & : & 4 \\ 0 & 3 & 6 & 9 & : & 12 \\ 0 & -2 & -4 & -6 & : & -8 \end{pmatrix} \underset{\substack{R_3-3R_2 \\ R_4+2R_2}}{\sim} \begin{pmatrix} \boxed{-1} & 1 & 1 & 2 & : & 7 \\ 0 & \boxed{1} & 2 & 3 & : & 4 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \end{pmatrix}.$$

where the second pivot at $a_{22} = 1$ gives us the echelon form, after only three row operations. Pivot columns are therefore columns 1 and 2, columns 3 and 4 are free. The rank is the number of pivot columns, so $\text{rank}(A) = 2$. The pivot columns of A are the basis of the column space:

Basis for $C(A)$:

$$(-1,0,2,0)^T \quad \text{and} \quad (1,1,1,-2)^T .$$

To find a basis for the nullspace we solve $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. Free variables are x_3 and x_4, and we begin by letting $x_3 = 1$ and $x_4 = 0$. In echelon form, we are left with two equations

$$-x_1 + x_2 + x_3 + 2x_4 = 0 \quad \text{and} \quad x_2 + 2x_3 + 3x_4 = 0$$

From the second equation, $x_3 = 1, x_4 = 0 \Rightarrow x_2 = -2$. Substitute these into the first equation and get $x_1 = -1$, giving the first special solution. Letting $x_3 = 0, x_4 = 1$, the second equation gives $x_2 = -3$ and substituting into the first equation, we get $x_1 = -1$ and the special solutions give the basis for the nullspace:

$$\text{basis for } N(A) \ : \ \underline{\mathbf{x}}_1^s \ = \ \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \underline{\mathbf{x}}_2^s \ = \ \begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

Now let the columns of A be $C_{1,2,3,4}$. For the basis vectors of the nullspace, we can write

$$A\underline{\mathbf{x}} = \underline{\mathbf{0}} \Rightarrow x_1C_1 + x_2C_2 + x_3C_3 + x_4C_4 \ = \ \underline{\mathbf{0}}$$

The special solutions which form the basis give the column dependence:
$-C_1 - 2C_2 + C_3 = 0 \Rightarrow C_3 = C_1 + 2C_2$ and $-C_1 - 3C_2 + C_4 = 0 \Rightarrow C_4 = C_1 + 3C_2$.

Finally, considering the last column of the echelon form, we get two equations:

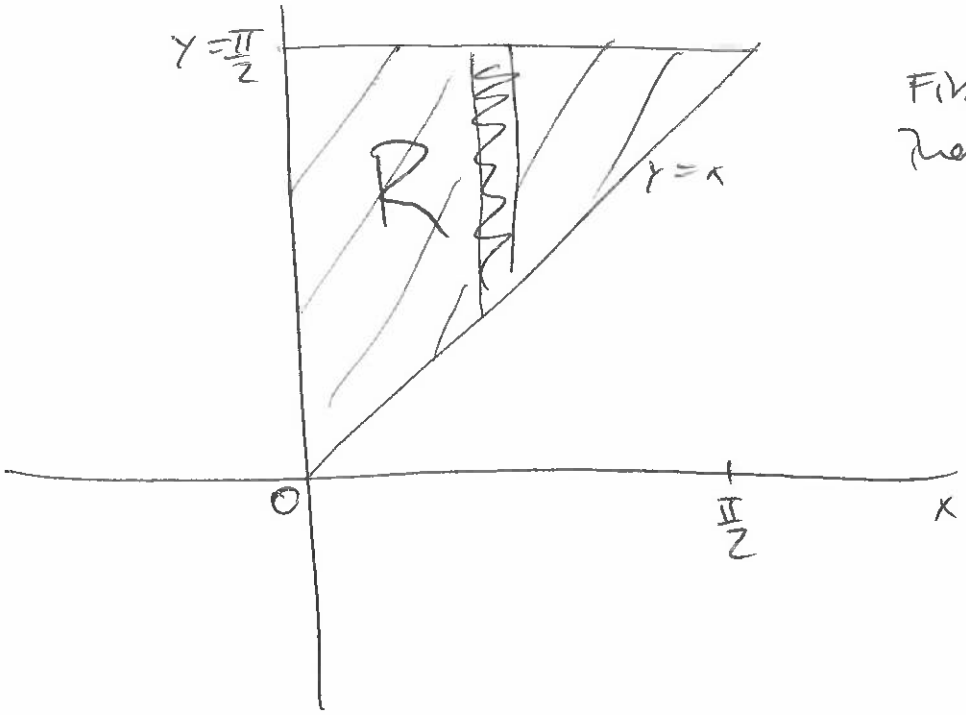
$$-x_1 + x_2 + x_3 + 2x_4 = 7 \quad \text{and} \quad x_2 + 2x_3 + 3x_4 = 4$$

and setting the free variables $x_3 = x_4 = 0$, the second equation gives $x_2 = 4$, and hence, the first equation has $x_1 = -3$, giving a particular solution $\underline{\mathbf{x}}_p = (-3, 4, 0, 0)^T$ and the general solution is obtained by taking the particular solution and the nullspace:

$$\underline{\mathbf{x}} \ = \ \begin{pmatrix} -3 \\ 4 \\ 0 \\ 0 \end{pmatrix} + p \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + q \begin{pmatrix} -1 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

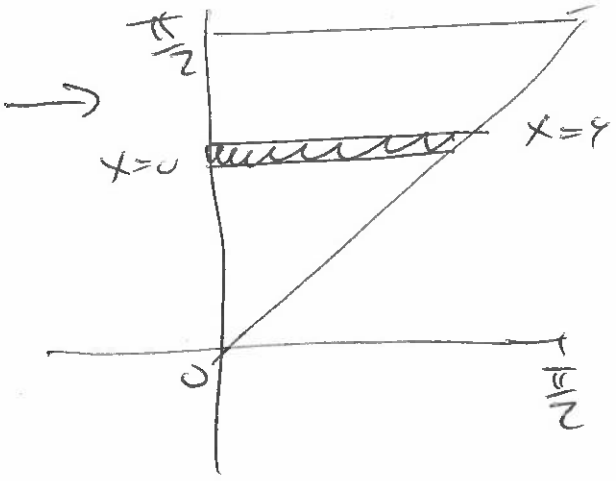
A lot of errors in row ops led to problems further on. Column space mostly right, though a few did use the columns from the echelon form. Nullspace usually fine, but a few used the equations from $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$, which introduced errors. Only a few people talked about what the basis for the nullspace says about the columns of A.

Q1(b)



First: $y = x \ldots \frac{\pi}{2}$

Then $x = 0 \ldots \frac{\pi}{2}$



First

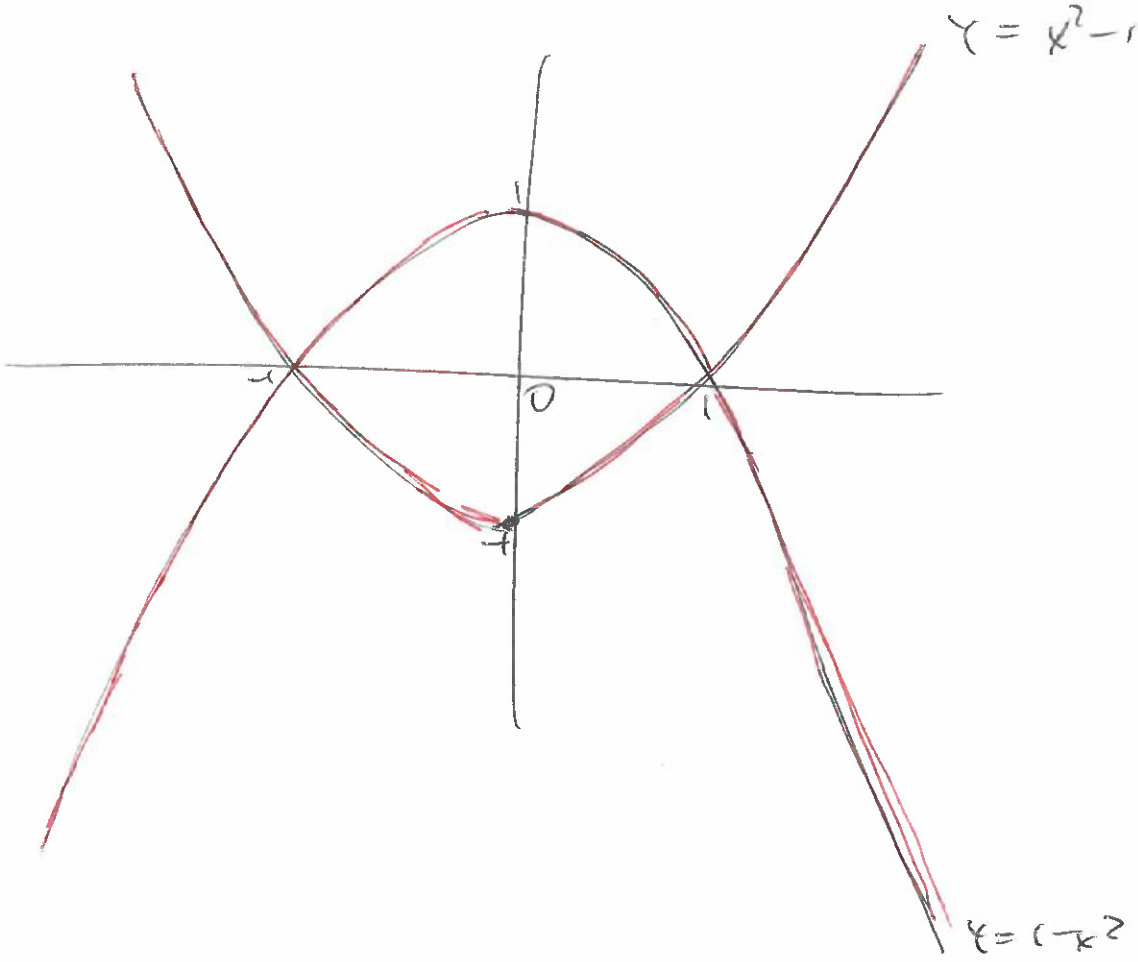
$x = 0 \ldots y$

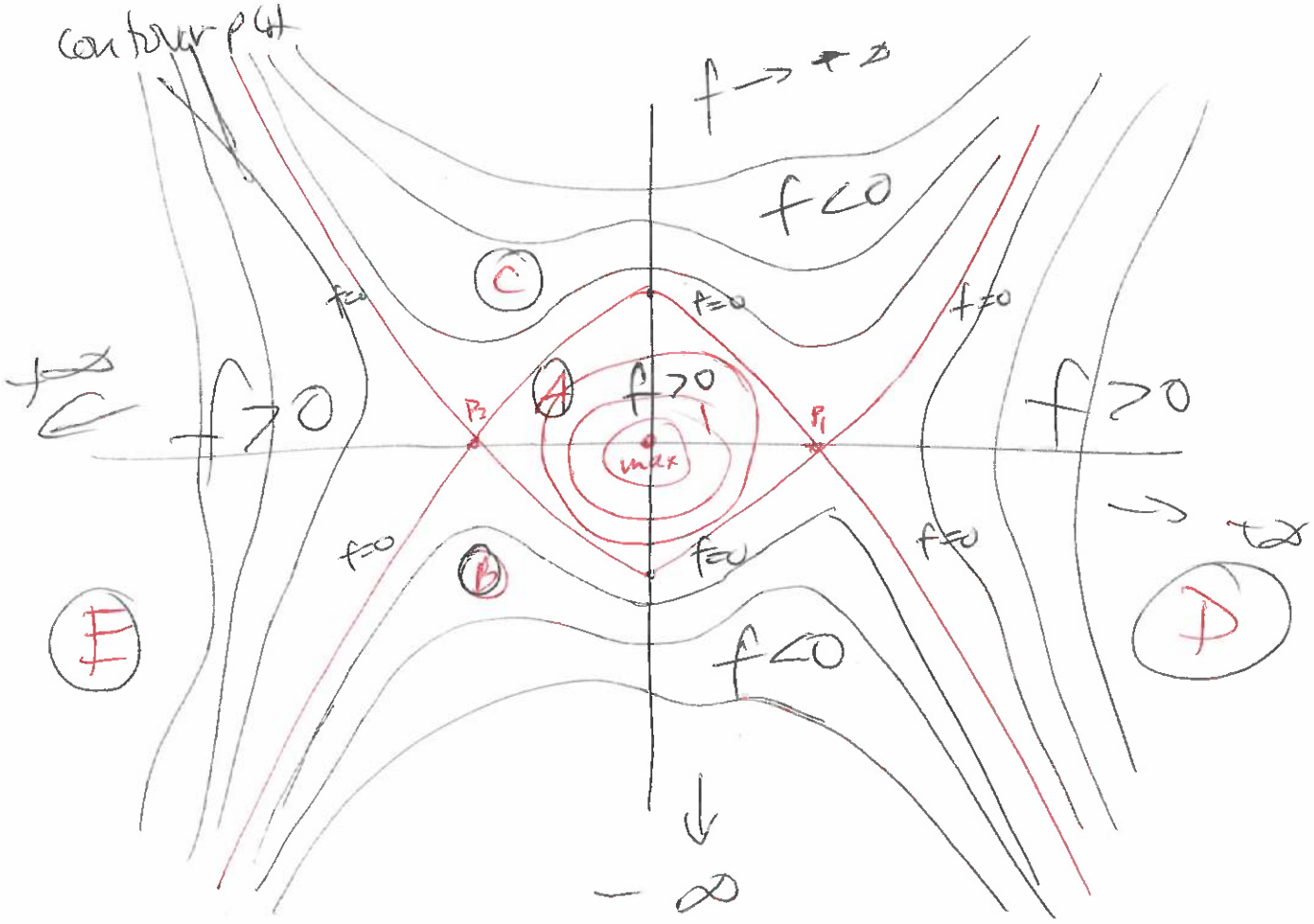
Then

$y = 0 \ldots \frac{\pi}{2}$

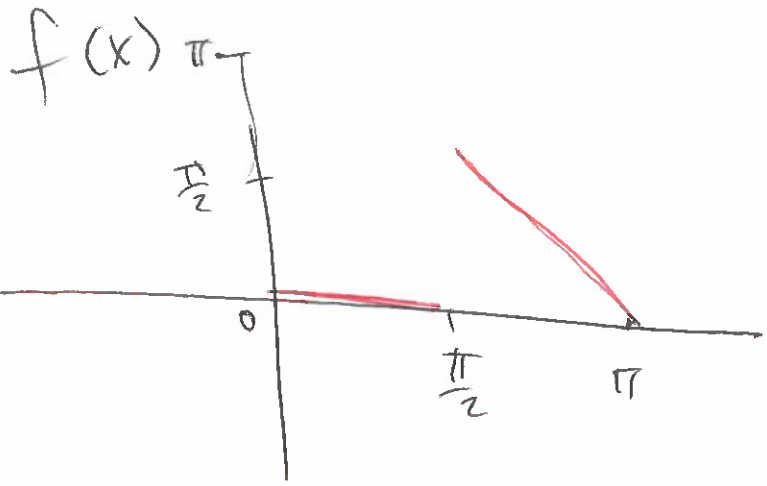
Q(c)

zero contour:

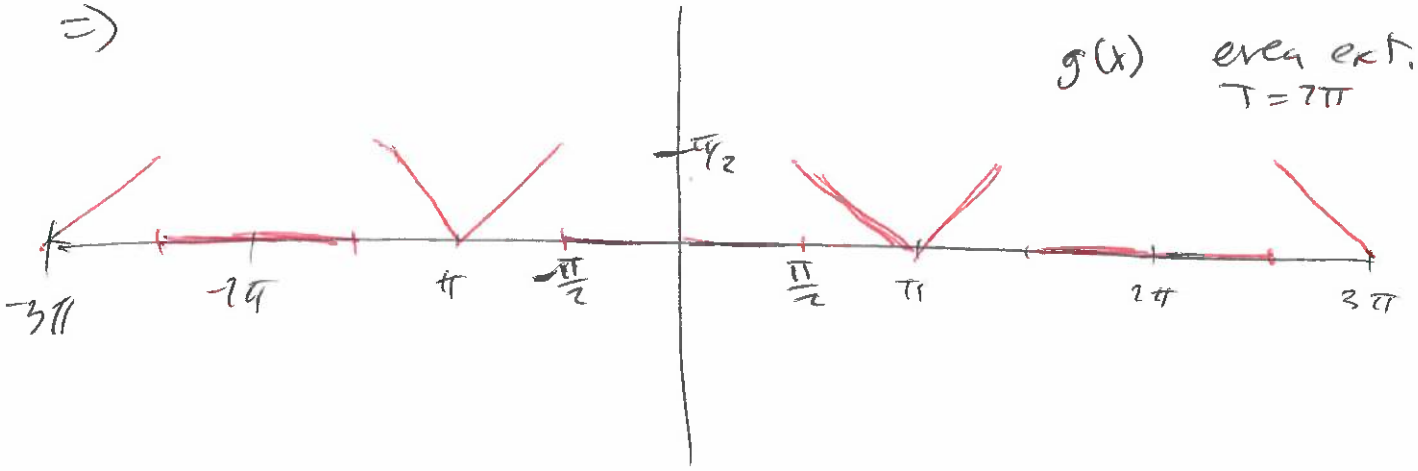


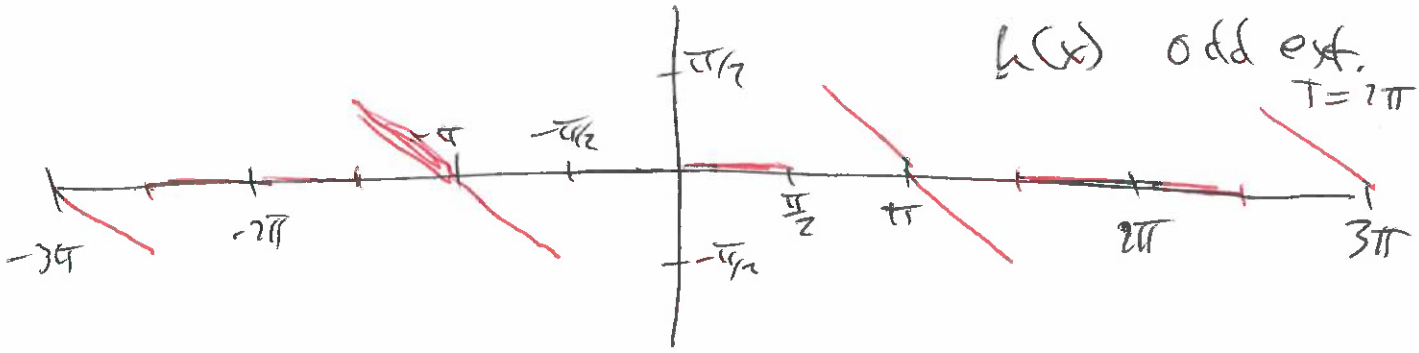


Q26


f(x)
π
π/2
0
π/2
π

⇒


g(x) even ext.
T = 2π
π/2
-3π
-2π
-π
-π/2
π/2
π
2π
3π


h(x) odd ext.
T = 2π
π/2
-3π
-2π
-π
-π/2
π/2
π
2π
3π